

Sub-Pixel Detection of Coastal Fish Cages and Kelongs

BUCKET 8 - COASTAL AND AQUACULTURE

For Singapore Food Agency (SFA)-licensed coastal fish farmers and Johor-side kelong operators in the Johor Strait — whose individual net cages and kelong platforms are physically smaller than a single Sentinel-2 pixel, so published pond/raft-extraction methods (U2-Net, HCHNet) have nothing to segment — this project reframes detection as sub-pixel density/count estimation fusing Sentinel-1 SAR (strong metal/floating-structure response) with Sentinel-2 multi-temporal persistence, validated against SFA's public licensed-farm register.

STRONG

This candidate passes all six hard constraints with a cleaner, more clearly bounded technical novelty argument than the companion bloom-forecasting candidate (existing methods provably fail at this scale, per the SAM-assisted-mariculture paper's own admission), and has a real, directly-verified authoritative farm register to validate against, even though that register needs manual geocoding work the student has not yet done.

Confidence: Medium-High. Confidence would rise to High with: (a) a week-1 confirmation that SFA/DOF farm codes can be geocoded to at least zone-level coordinates, and (b) a quick visual check that farm sites are actually distinguishable in real Sentinel-1/2 imagery over the Johor Strait. It would fall if either check fails, since the entire validation strategy depends on having some geocoded reference, even coarse.

Hard constraint check

Constraint	Verdict	Reasoning
H1 · Novelty	PASS	existing pond/raft/cage extraction methods (U2-Net, HCHNet, AF-UNet++, SAM-assisted) operate at a coarser structural scale and, even there, are reported to struggle at fine raft/cage granularity; the sub-pixel density/count reformulation for Johor Strait's genuinely smaller structures is a real technical problem created by the geography, not a re-run of an existing pipeline.
H2 · Not generic	PASS	not any of the 8 banned patterns; the one-sentence pitch is specific to the sub-pixel regime, which existing generic aquaculture-mapping FYPs would not naturally describe.
H3 · Imagery is primary input	PASS	Sentinel-1 SAR and Sentinel-2 optical imagery are the sole primary inputs.
H4 · Geography & user	PASS	named user is SFA-licensed and Johor DOF/myGAP-licensed coastal fish farm and kelong operators in the Johor Strait, a real SEA, Singapore-preferred user segment.
H5 · Compute & budget	PASS	smaller compute and data-volume footprint than the companion bloom-forecasting candidate; comfortably fits MacBook Pro M4 Pro 24GB unified memory without needing Colab/Kaggle fallback, and RM0 core cash cost.

named user and a technically non-trivial, demoable core; quantified pain is weaker than ideal (no single number for losses this system specifically addresses was found this session) and should be strengthened before the panel pitch.

The problem

Coastal aquaculture mapping from satellite is a mature research area for China's and Southeast Asia's large raft, cage-cluster and pond systems, which are extracted at 10-16m resolution with high accuracy (U2-Net F-measure 0.93 on Zhoushan Archipelago ponds; HCHNet >95% accuracy nationally in China). But Johor Strait and Singapore coastal fish farms are structurally different: SFA's own published farm typology describes coastal farming SITES of about 0.5-1 ha (roughly 70-100m per side, i.e. 7-10 Sentinel-2 pixels across), but the individual net cages and kelong (stilted platform) structures WITHIN a site are typically on the order of 10-20m or less — at or below a single 10m Sentinel-2 pixel, and well below what any published pond/raft segmentation method was designed or validated to resolve. This means the unit of interest (an individual cage or kelong) frequently falls at or below the sensor's nominal resolution, so a segmentation formulation (draw a boundary around each object) is the wrong formulation: there may be no boundary to draw in a 10m pixel. Existing published methods (U2-Net, HCHNet, AF-UNet++, SAM-assisted mariculture extraction) either target much larger raft/pond footprints, or — when applied to raft/cage-scale structures — are explicitly reported to fail without manual interpretation (Ren et al. 2023 report SAM-assisted raft extraction as 'challenging and remains unsatisfactory').

Proposed technical approach

Reframe the task from instance/semantic segmentation to sub-pixel density/count estimation, following the crowd-counting and remote-sensing-object-counting literature (density-map regression rather than bounding-box/mask prediction). Inputs: Sentinel-1 GRD dual-pol (VV/VH) backscatter time series — floating metal/timber structures and calm strait water produce a strong, temporally-persistent backscatter contrast even when individual structures are sub-pixel, because SAR backscatter is sensitive to structure presence/orientation rather than requiring a fully-resolved outline — fused with Sentinel-2 multi-temporal composites (persistent water-color/texture anomalies at farm-site locations across many cloud-free dates, exploiting the fact that a farm site persists for years while clouds and glint vary date to date). Architecture: a small CNN (e.g. a lightweight density-estimation head, in the style of MCNN/CSRNet-family crowd-counting networks adapted to remote-sensing object counting, c.f. Jiang et al. 2023 arXiv: 2303.10318) trained to regress a per-pixel density map whose integral over a farm-site polygon equals the estimated cage/kelong count, rather than a mask whose IoU is scored per-instance. Training signal: weak/coarse supervision from farm-site locations (approximate, since SFA's public register lists farm codes and licensees but not precise lat/lon — see datasets/verification below) combined with self-supervised multi-temporal persistence (a real farm site produces a stable multi-date SAR+optical anomaly; noise does not), plus a small manually-photo-interpreted validation set (using free high-resolution basemap imagery, e.g. Google/Bing satellite basemaps or ESA WorldCover-adjacent VHR previews, for visual cage-counting at a handful of known farm sites) to calibrate and validate the density-to-count mapping. Output: a per-farm-zone estimated cage/kelong density and count, cross-checked against the SFA licensed-farm register's East/West Johor Strait farm-code convention.

Novelty

WHAT THE DELTA ACTUALLY IS

The defensible delta is a new data/method regime, not a new region alone. Published coastal aquaculture extraction methods (U2-Net F=0.93, HCHNet >95% accuracy, AF-UNet++, SAM-assisted extraction) are validated on China's raft/pond/cage systems, which are large enough relative to 10-16m pixels that

segmentation is a well-posed formulation — indeed, even in that easier regime, the SAM-assisted paper (Ren et al. 2023) explicitly reports raft-scale extraction as still unsatisfactory without manual interpretation. Johor Strait/Singapore cages and kelong are smaller still relative to the individual-structure unit of interest, placing them below the resolution regime any of these published methods were designed for. This is the H1 'genuine technical problem created by geography' case explicitly, not 'applied X to Y country': the segmentation formulation itself breaks down (there may be no resolvable boundary in a 10m pixel), which is why the method must change to density/count estimation, a different problem formulation borrowed from the crowd-counting/remote-sensing-object-counting literature (c.f. arXiv:2303.10318) and applied to a sub-pixel structure regime that, to the depth of this session's search, no existing aquaculture-mapping paper addresses.

PRIOR ART THAT THREATENS IT

No 2024-2026 paper was found in this session's arXiv/OpenAlex/DBLP searches that performs sub-pixel density/count-based detection of small floating aquaculture structures (cages/kelongs) using SAR+optical fusion. The closest adjacent papers (AF-UNet++ 2024 for cage/raft segmentation at China scale; SAM-assisted mariculture extraction 2023, which explicitly fails at fine raft/cage scale) confirm rather than threaten the novelty claim, since both operate in the coarser regime this idea explicitly argues is a different, easier problem. The main residual threat is not a specific paper but a generic one: general remote-sensing small-object-counting/density-estimation methods (e.g. crowd counting adapted to satellite imagery, arXiv:2303.10318) already exist as a technique family, so the contribution here is the APPLICATION of that technique family to the aquaculture sub-pixel regime plus SAR-optical fusion specifically, not the invention of density estimation itself — this must be stated honestly rather than oversold as a brand-new method.

GENERIC-PATTERN CHECK

Core is sub-pixel density/count estimation via SAR+optical fusion for structures below segmentation resolution — not a chatbot/LLM assistant, not a dashboard-as-core, not leaf-photo disease classification, not an NDVI viewer, not tabular yield-from-CSV prediction, and explicitly **NOT** a re-run of an existing crop/pond-classification benchmark (the whole premise is that the existing benchmark formulation, segmentation, does not apply at this scale). Could the one-sentence summary describe 50 other FYPs? No — 'sub-pixel density estimation for small floating aquaculture structures using SAR+optical fusion, validated against a licensed-farm register' is narrow and specific; the closest generic pattern it could be confused with (coastal aquaculture pond mapping) is explicitly the thing it argues does **NOT** apply here.

Sensors and imagery

Sensor	Resolution	Revisit (days)	Bands	Licence
Sentinel-1 GRD (IW mode)	10 m	6-12	VV, VH dual-pol backscatter	Free, open (Copernicus)
Sentinel-2 MSI	10 m	~5	Blue, Green, Red, NIR (multi-temporal water/structure persistence composites)	Free, open (Copernicus)

Two-sensor fusion (SAR + optical); both satellite, both primary inputs. No drone/UAV imagery used in this candidate.

Yes. Sentinel-1 SAR backscatter and Sentinel-2 multi-temporal optical composites are the sole model inputs; there is no non-imagery primary input.

Datasets

SENTINEL-1 GRD (COPERNICUS DATA SPACE ECOSYSTEM / ASF VERTEX)

Access route	dataspace.copernicus.eu or search.asf.alaska.edu, free self-registration
Auth required	Free account registration
Licence	Copernicus free and open data policy
Approx. size	Estimated 10-20GB for a multi-year, AOI-cropped IW GRD stack over the narrow Johor Strait
Accessible from MY	True
How verified	WebSearch confirmed CDSE/ASF free global access, no region restriction found

SENTINEL-2 L2A (COPERNICUS DATA SPACE ECOSYSTEM)

Access route	dataspace.copernicus.eu, free self-registration, STAC/OData API
Auth required	Free account registration
Licence	Copernicus free and open data policy
Approx. size	Estimated 10-20GB for AOI-cropped multi-temporal composites
Accessible from MY	True
How verified	WebSearch confirmed CDSE global free access

SFA LIST OF LICENSED SEA-BASED FARMS (SINGAPORE FOOD AGENCY)

Access route	https://www.sfa.gov.sg/docs/default-source/food-farming/licensed-sea-based-farms.pdf — fetched and parsed directly this session
Auth required	None, public PDF
Licence	Public government publication
Approx. size	<0.001
Accessible from MY	True
How verified	Directly downloaded and text-extracted this session (pypdf) — confirmed real, current document, 'Last updated 03 July 2026', listing 68 licensed sea-based farm entries with farm codes (e.g. FC(I)108E, FC12W, NE5C, NW12C, SJ11) and licensee names. IMPORTANT VERIFIED FINDING: the register gives farm codes, licence numbers and operator names, and the code suffix (E/W/C) plausibly maps to East/West Johor Strait or Southern-islands zones per SFA's own descriptive text (West Johor Strait off Lim Chu Kang; East Johor Strait off Seletar/Pasir Ris/Pulau Ubin/Pulau Tekong), but the PDF itself contains NO latitude/longitude coordinates or geocoded farm-site polygons. It is an authoritative identity/licensing register, not a geocoded location dataset — manual geocoding (e.g. against SFA's individual farm-profile pages at sfa.gov.sg/fromSGtoSG/farms , or against the known farming-site zones) would be a required, non-trivial week-1/2 engineering task, not something to assume is solved.

MALAYSIA DOF MYGAP-CERTIFIED AQUACULTURE PREMISES LIST (JOHOR)

Access route	dof.gov.my/en/services/certification-scheme/list-of-mygap-certified-aquaculture-premises/
Auth required	None, public listing
Licence	Public government publication
Approx. size	<0.001
Accessible from MY	True
How verified	[uncertain] page existence confirmed via WebSearch; the list was not directly fetched/parsed this session, so whether it includes Johor Strait coastal (vs freshwater/onshore) premises with any location detail is not yet confirmed

HIGH-RESOLUTION BASEMAP IMAGERY FOR VISUAL CAGE-COUNTING CALIBRATION (E.G. GOOGLE/BING SATELLITE BASEMAP PREVIEWS)

Access route	Free web-based basemap viewing for manual visual interpretation only (not bulk download); this is NOT a licensed commercial VHR imagery dependency, only used for small-N manual labelling
Auth required	None for casual viewing
Licence	Free viewing under standard basemap terms; not redistributable, used only for manual label creation, not as training input to the model
Approx. size	<0.01 (screenshots for manual annotation only)
Accessible from MY	True
How verified	General knowledge; not independently re-verified this session, flagged as [uncertain] whether current Johor Strait basemap imagery resolution is sufficient to visually count individual cages

TOTAL DATA VOLUME

Estimated 20-40GB using cloud-native cropped access (STAC/COG range reads over the narrow Johor Strait AOI) for the Sentinel-1+Sentinel-2 stack; well under the ~200GB H5 practical ceiling given the small two-sensor, no-OLCI footprint of this candidate compared to the bloom-forecasting candidate.

LABELS AND GROUND TRUTH

Ground truth is a genuine, verified constraint. The SFA licensed-farm register (directly parsed this session, 68 entries) is authoritative for WHICH farms are licensed and gives a coarse East/West/zone signal via farm codes, but has **NO** coordinates — it cannot alone serve as a geocoded label set. Substitutes proposed: (1) manual geocoding of the ~68 SFA farm codes against SFA's published farming-site zone descriptions and individual farm-profile pages (a real, bounded engineering task, consistent with the S2 build-heavy preference, but not yet done — flagged as a week-1/2 task); (2) small-N manual photo-interpretation of individual cage/kelong counts at a handful of geocoded sites using free high-resolution basemap imagery, to create a limited but real calibration/validation set for the density-to-count regression (this is the standard approach in remote-sensing object-counting literature when exhaustive ground truth is unavailable); (3) self-supervised multi-temporal persistence as the main TRAINING signal (real farm sites recur at the same location across many SAR/optical dates; this requires no manual labels at all and only needs the coarse farm-site locations from (1) for evaluation, not training). Risk carried: the coordinate-geocoding step for both the Singapore (SFA) and Malaysia (Johor DOF) sides is unverified this session and

could be more laborious than assumed if SFA's individual farm-profile pages do not expose usable location data either.

Baselines to beat

Paper	Reported metric	Dataset
Extraction of Aquaculture Ponds along Coastal Region Using U2-Net Deep Learning Model from Remote Sensing Images Remote Sensing (MDPI) 2022 · 10.3390/rs14164001	Average F-measure 0.93, precision 92.21%, recall 93.79%	Zhoushan Archipelago, China, coastal aquaculture ponds
A new satellite-derived dataset for marine aquaculture areas in China's coastal region (HCHNet) Earth System Science Data 2021 · 10.5194/essd-13-1829-2021	Overall classification accuracy >95% (95.2-96.4%, 95% CI)	China GF-1 satellite, 16m resolution, national-scale marine aquaculture
An Attention-Fused Deep Learning Model for Accurately Monitoring Cage and Raft Aquaculture at Large-Scale Using Sentinel-2 Data (AF-UNet++) IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS) 2024 · 10.1109/jstars.2024.3390762	Highest OA/F1/mIoU among compared CNNs; IoU for raft class 4.15% higher than other models	China offshore seas, Sentinel-2, cage and raft aquaculture, 32x32 patches
Segment Anything Model (SAM) Assisted Remote Sensing Supervision for Mariculture — Using Liaoning Province, China as an Example Remote Sensing (MDPI) 2023 · 10.3390/rs15245781	Cage mariculture extraction efficient via SAM; raft mariculture extraction explicitly reported as 'challenging and remains unsatisfactory... manual interpretation still required'	Liaoning Province, China, high-resolution imagery
Remote Sensing Object Counting with Online Knowledge Learning arXiv preprint 2023 · arXiv: 2303.10318	Density-map-based counting framework for remote sensing objects (methodological baseline for the density-estimation reformulation this idea proposes, not aquaculture-specific)	General remote-sensing object-counting benchmarks

Evaluation protocol

Primary metric: count error (mean absolute error / mean absolute percentage error) between predicted density-map integral and manually-photo-interpreted ground-truth cage/kelong counts at a held-out set of geocoded farm sites. Secondary metric: farm-site-level presence/absence detection F1 (does the model correctly flag a farm-site zone as containing structures at all), evaluated against the SFA 68-entry register at the zone level (since individual-cage-level ground truth will be small-N). Geographic generalisation: train on West Johor Strait farm codes, evaluate on East Johor Strait farm codes (or vice versa), since SFA's own documentation states these zones farm different species under different site conditions (West: milkfish/grey mullet/tilapia in shallower, lower-oxygen sites off Lim Chu Kang; East: higher-value Asian seabass/grouper/snapper off Seletar/Pasir Ris/Pulau Ubin/Pulau Tekong), a genuine domain-shift test. Success would be defined as: farm-site presence/absence F1 that clearly beats a naive water-texture-anomaly baseline, and a count-error that is at least directionally informative (e.g. correctly rank-orders farm sites by structure density) even if absolute count accuracy is modest given the sub-pixel regime.

Feasibility

COMPUTE

Peak memory	Estimated 4-8GB for a lightweight density-estimation CNN (MCNN/CSRNet-style) at small patch sizes, comfortably within 24GB unified memory
Est. training hours (MPS)	Estimated 2-5 hours on MPS given the smaller model and smaller effective dataset (two sensors, small AOI) than the companion bloom-forecasting candidate
Fits 24 GB	True
MPS-compatible ops	Standard convolutional density-estimation architectures use ops well-supported on PyTorch MPS; no custom CUDA kernels expected
Colab/Kaggle fallback	False
Notes	This is a lighter compute profile than the companion bloom-forecasting candidate; the dominant cost is expected to be data engineering (SAR preprocessing, multi-temporal compositing, and especially manual geocoding of the SFA farm-code register) rather than model training.
Cash cost	Estimated RM0 core cost: Sentinel-1/2 data are free (Copernicus Data Space Ecosystem), the SFA register is a free public PDF, and basemap imagery for small-N manual calibration labelling is free to view (not bulk-licensed). No commercial VHR imagery dependency. Full RM200 budget remains available as buffer.
Build : research	Approximately 75:25 engineering-to-research. The density-estimation reformulation itself is a known technique family (light research/positioning work); the bulk of the effort is engineering: SAR/optical data pipeline, multi-temporal compositing, manual geocoding of the SFA/DOF farm registers, and a small manual photo-interpretation labelling pass — matching the S2 build-heavy preference well.
Publication venue fit	IGARSS or Remote Sensing (MDPI) fit well — both venues regularly publish coastal aquaculture remote-sensing mapping papers (see named_baselines, several of which are exactly this venue pair), and a sub-pixel density-estimation reformulation for small structures is a natural, publishable methodological contribution to that existing literature.

DEMO ARTIFACT

Live on a laptop: select a stretch of the Johor Strait containing several SFA farm codes; show the raw Sentinel-1 SAR and Sentinel-2 optical multi-temporal composite side by side, then the model's predicted density heatmap overlaid on the same stretch, with estimated counts per farm-code zone displayed next to the SFA register's East/West zone labels for a direct, visual sanity check the panel can follow in under 2 minutes.

User and pain

USER SEGMENT

SFA-licensed coastal sea-based fish farm operators in the Johor Strait (68 currently-licensed entries confirmed by direct parsing of SFA's public register this session; SFA's own published figures state ~110 sea-based farms in coastal and southern waters contributing ~85% of Singapore's local foodfish production) and Johor Department of Fisheries-licensed / myGAP-certified marine fish farm and kelong operators on the Malaysia side of the strait. Output plausibly also useful to SFA/Johor DOF as a regulatory compliance/monitoring aid (detecting unlicensed or expanding structures), though the core deliverable targets the farmer/regulator pair named here, not a generic monitoring-portal audience.

QUANTIFIED PAIN

SFA sea-based coastal farms contribute approximately 85% of Singapore's local foodfish production from about 110 farms (source: World Aquaculture Society summary of Singapore aquaculture industry, corroborated by SFA's own public farm-typology page). **[uncertain: the 110-farm figure from secondary sources is somewhat higher than the 68 entries directly counted in the currently-in-force SFA licensed-farms PDF parsed this session — this discrepancy (68 vs ~110) was not reconciled this session and should be resolved by the student, e.g. by checking whether the 110 figure includes closed-containment/other non-net-cage farm types not on the 'sea based' list, or reflects an earlier/different count date]** There is no single quantified number this session found for economic loss specifically attributable to UNDETECTED illegal/unlicensed cage expansion or storm/collision damage that this detection system would address, distinct from the algal-bloom fish-kill losses quantified in the companion candidate — this is a genuine gap in the quantified-pain argument that the student should shore up, e.g. via SFA/DOF compliance reports if accessible.

Key risks

1. Geocoding risk (highest, data-access risk): the SFA register (verified this session) has no coordinates; if SFA's individual farm-profile pages and/or Johor DOF records also lack usable location data, building even a coarse geocoded validation set could consume disproportionate project time.
2. Ground-truth risk: individual-cage-level ground truth requires manual photo-interpretation from basemap imagery whose current resolution over the Johor Strait was not verified this session — if basemap imagery is too coarse or outdated to visually count cages, the calibration/validation set cannot be built as proposed.
3. Method risk: SAR backscatter response from small floating structures against calm strait water is well-established qualitatively (this is why ship/vessel SAR detection works, c.f. arXiv:2507.04842 xView3-SAR-based work), but whether it holds for the smaller, lower-metal-content kelong (often timber-stilted) structures specifically was not verified against a named prior study this session.
4. Evaluation risk: with likely small-N geocoded validation sites, count-error metrics may have high variance; the evaluation protocol should be designed to also report a coarser, more robust presence/absence metric as a fallback.

5. Quantified-pain gap: no single number for economic loss from undetected/unlicensed structure changes was found this session, weakening the H6 pitch relative to the companion bloom candidate, which has a concrete historical loss figure.

Kill criteria

1. Week 1: attempt to geocode a sample of 10-15 SFA farm codes against SFA's individual farm-profile pages or any other public source; if no usable location data can be recovered beyond the coarse East/West/zone signal already in the farm code, the validation approach must be redesigned (e.g. relying entirely on zone-level rather than site-level evaluation).
2. Week 1: pull one Sentinel-1 and one Sentinel-2 scene over a known farm-dense stretch of the Johor Strait and visually confirm that farm sites are actually visible as a distinguishable backscatter/optical anomaly at all; if farm sites are indistinguishable from background water/noise even to a human eye at 10m resolution, the sub-pixel detection premise itself may not be viable and the idea should be abandoned or re-scoped to farm-SITE (not individual-cage) detection only.
3. Week 2: check current basemap imagery resolution/recency over the Johor Strait; if individual cages cannot be visually counted from any accessible free imagery, the manual calibration/validation step cannot be built as designed.
4. Week 2: confirm whether Johor DOF/myGAP records include any Malaysia-side Johor Strait marine cage sites with usable location info; if the Malaysia side yields nothing, the project may need to narrow scope to the Singapore side only, which weakens the SEA/regional framing but does not by itself disqualify the idea.

Open / unverified items

- datasets[myGAP list]: not directly fetched/parsed this session; whether it includes Johor Strait coastal (as opposed to freshwater/onshore) premises with any location detail is unconfirmed
- datasets[basemap imagery]: current resolution/recency of free basemap imagery over the Johor Strait, sufficient for manual cage-counting, was not independently verified this session
- quantified_pain: the discrepancy between the 68 entries directly counted in the current SFA licensed-farms PDF and the ~110-farm figure quoted in secondary sources was not reconciled this session
- quantified_pain: no single quantified economic-loss figure specific to undetected/unlicensed structure changes (as opposed to bloom-driven fish-kill losses) was found this session
- key_risks[3]: whether SAR backscatter response holds specifically for timber-stilted kelong structures (as opposed to metal-hulled vessels, where the SAR response is well-established) was not verified against a named prior study this session
- labels_and_ground_truth: whether SFA's individual farm-profile pages (sfa.gov.sg/fromSGtoSG/farms) expose usable location/coordinate data was not directly fetched and checked this session

Key references

Extraction of Aquaculture Ponds along Coastal Region Using U2-Net Deep Learning Model from Remote Sensing Images

Remote Sensing (MDPI) · 2022 · [10.3390/rs14164001](https://doi.org/10.3390/rs14164001)

Named in the candidate description as an existing method with 'nothing to segment' at Johor Strait's scale; establishes the coarser regime ($F=0.93$ at China's larger pond scale) this idea must explicitly contrast against.

A new satellite-derived dataset for marine aquaculture areas in China's coastal region (HCHNet)

Earth System Science Data · 2021 · [10.5194/essd-13-1829-2021](https://doi.org/10.5194/essd-13-1829-2021)

The second named baseline in the candidate description; a national-scale (16m resolution) segmentation approach whose scale assumptions do not hold for Johor Strait's smaller structures.

Segment Anything Model (SAM) Assisted Remote Sensing Supervision for Mariculture — Using Liaoning Province, China as an Example

Remote Sensing (MDPI) · 2023 · 10.3390/rs15245781

Directly corroborates the novelty claim: even a state-of-the-art foundation-model-assisted segmentation approach (SAM) is reported to fail at fine raft-scale extraction, requiring manual interpretation — strong evidence that segmentation is the wrong formulation at small scales.

An Attention-Fused Deep Learning Model for Accurately Monitoring Cage and Raft Aquaculture at Large-Scale Using Sentinel-2 Data

IEEE JSTARS · 2024 · 10.1109/jstars.2024.3390762

The most recent and closest published cage/raft-specific Sentinel-2 method; a required direct comparison point since it targets the same object classes (cage, raft) this idea targets, just at a larger, China-specific scale.

Remote Sensing Object Counting with Online Knowledge Learning

arXiv preprint · 2023 · arXiv:2303.10318

Establishes the density-estimation/object-counting technique family this idea borrows and applies to the aquaculture sub-pixel regime; important for correctly scoping the novelty claim as an application/adaptation rather than a new base method.

Generated from the source proposal JSON. Amber [uncertain] flags are carried over verbatim from the source and mark claims that were not independently verified.